

RIDDHI KULKARNI

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OBJECTIVE

To pursue an MS in Artificial Intelligence and Machine Learning to deepen my expertise in advanced machine learning, deep learning architectures, and large-scale model deployment while mastering the full lifecycle of intelligent systems, from model formulation and learning to deployment and evaluation at scale. Post-graduation, I seek to work as an Applied AI/ML Engineer designing end-to-end learning systems that translate research-grade models into reliable real-world applications.

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Telecommunications Engineering,

Pimpri Chinchwad College of Engineering (PCCOE), Pune

November 2022 – June 2026

CGPA: 8.23

Additional Honors Degree:

Deep Learning (Computer Science Department) – CGPA: 8.5

July 2024 - June 2026

PROJECTS

Wriddles: AI-Powered Gamified Learning Platform

June 2025 – August 2025

Asymmetrical.ai, Pune

Team Size: 2

Role: Team Lead/AI/Backend Developer

Description:

- Developed backend intelligence for Wriddles, an AI-powered gamified learning application sponsored and published by Asymmetrical.ai on the Google Play Store.
- Implemented a reinforcement learning-based personalization engine using the LinUCB algorithm to dynamically adjust riddle difficulty and recommendations.
- Built and deployed 40+ backend services using Python, Flask, Firebase, and Google Cloud Functions to support scalable adaptive learning.
- Enabled adaptive content delivery and personalized learning using reinforcement learning-based recommendation.

AI-Based Smart Water Quality Analyzer & Disease Outbreak Forecasting System

Pimpri Chinchwad College of Engineering, Pune

June 2025 – November 2025

Team Size: 3

Role: Team Lead/AI/Backend Developer

Description:

- Designed an IoT-enabled water quality monitoring system using ESP32 sensors to track pH, turbidity, dissolved oxygen, and temperature.
- Developed four machine learning models including Random Forest and bidirectional LSTM for WQI prediction ($R^2 = 0.95$) and contamination detection (97.7% accuracy).
- Constructed a dataset of 48k+ sensor readings using noise augmentation and temporal slicing for robust training.
- Built a real-time dashboard to visualize sensor data, predictive analytics, and early-warning alerts for water safety monitoring.

Multi Modal Transformer for Cell Type Specific Chromatin Accessibility using DNA sequence and functional genomic signals.

Pimpri Chinchwad College of Engineering, Honors-Deep Learning, Pune

June 2025 – November 2025

Team Size: 4

Role: Research and Development, Data Collection and Preprocessing.

Description:

- Preprocessed large-scale ENCODE datasets using pyBigWig, pybedtools, and pyfaidx to generate fused multimodal inputs.
- Implemented cross-modal attention mechanisms and trained the model using masked-accessibility objectives in PyTorch.
- Achieved improved ROC-AUC and PR-AUC compared to sequence-only baselines while enabling interpretable motif discovery.

INTERNSHIPS

AI Intern, Asymmetrical.AI(Acceler EdTech Pvt. Ltd.), Pune, India

June 2025 – August 2025

- Developed backend architecture for *Wriddles*, an AI-powered gamified learning app built from scratch and published on the Google Play Store.
- Developed a PDF document understanding pipeline using OpenAI APIs and prompt engineering, achieving ~95% extraction accuracy.
- Collaborated and assisted in content writing for *The Asymmetrical Booktern* innovation magazine.

TECHNICAL SKILLS

Programming Languages: Python (4 years), C++ (3 years), Java (1 year), MATLAB (3 years), C (4 years)

Technical Skills: Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Computer Vision, Advanced Learning Algorithms

Frameworks and Tools: PyTorch, TensorFlow, Flask, Firebase, NumPy, Pandas

CERTIFICATIONS/WORKSHOPS/ADDITIONAL COURSES:

- **Edge AI Workflow Design (6-Day Workshop)**, Pimpri Chinchwad College of Engineering (PCCOE) & DigiToad, **Jul 2025**
- **Machine Learning**, Columbia University, edX, **Jul 2025 (Expected)**
- **Microsoft Azure AI Essentials Professional Certificate**, Microsoft, Microsoft Learn, **Jun 2025**
- **Python Programming Masterclass**, Udemy, **May 2025**
- **Machine Learning Onramp**, MathWorks (MATLAB), **Feb 2025**
- **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI, Coursera, **Dec 2024**
- **Object-Oriented Programming Using C++**, Coursera, **Mar 2023**

RESEARCH/PUBLICATIONS

Kulkarni, R., Gangrethiwar, M., Kshirsagar, A., Kulkarni, V., & Gaadhe, A.,
“AI Based Smart Water Quality Monitoring and Disease Outbreak Forecasting System,”
2025 IEEE Pune Section International Conference (PuneCon), Pune, India, Dec 2025. (Published)

Manuscripts Under Review :

Kulkarni, R., et al.,
“**Multi-Modal Transformer for Chromatin Accessibility Prediction**,”
Submitted to IEEE Conference, 2026. (*Under Review*)

Kulkarni, R., et al.,
“**AI-Powered Sleep Stage Analysis and Sleep Score Recommendation System**,”
Submitted to IEEE Conference, 2026. (*Under Review*)

ACHIEVEMENTS/CO-CURRICULAR/EXTRACURRICULAR

- Smart India Hackathon – National Level Event– **Round 2 Shortlisted**, Ministry of Education (MIC-AICTE), **2025**
- SPECTRUM State-Level Tech Event – **Finalist**, **2023**
- Navamal Firodiya National Debate Competition – **Finalist**, **2023**

LEADERSHIP ROLES

- Led a team of 30 students and provided advisory support in treasury management and organization of the national-level technical fest “NEXUS” as Overall Advisor, Electronics and Telecommunication Students’ Association (ETSA), Pimpri Chinchwad College of Engineering, 2025–2026.
- Organized and managed 2 technical events, 2 non-technical events, and 15+ workshops involving 350+ student participants as Secretary, Electronics and Telecommunication Students’ Association (ETSA), Pimpri Chinchwad College of Engineering, 2024–2025.
- Coordinated the Abhyantrix’25 Research Paper Conference for the entire ENTC department with participation from 200+ students as Overall Coordinator, Pimpri Chinchwad College of Engineering, 2025.

LANGUAGES KNOWN

English (proficient), Marathi (fluent), Hindi (fluent), and German (basic).